

ECE 198
Design Document
Group 16

Plant Care System
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1. Needs Assessment

1.1. Client Definition

1.1.1. Introduction

An increasing population of working young adults in Waterloo struggles to maintain plants due to watering, soil, light, and temperature challenges. As such, plants wilt or grow at stunted rates. 14% of plants die within their first year of life [1]. This is a discouraging sentiment for plant owners. A survey that interviewed 2,000 millennials found that 48% of interviewees were unsure about whether they could maintain the livelihood of their plants. 67% of individuals found “plant parenthood” to be tougher than initially presumed. Alongside these struggles comes the recognition that 81% of those surveyed were physically and mentally healthier upon integrating plants into their living arrangements [2].

Increasing developments in technology allow our target group to remain connected indoors. However, physiological stressors are associated with the lack of a natural environment. Indoor plants enhance job satisfaction in the workplace, reduce stress and improve one’s sentiments, and facilitate cognitive behaviours. As such, young adults who take the time to disengage from their computers notice a change in mood from “uncomfortable” to “soothed” and “natural” [3]. Waterloo Gardeners refers to gardening as a “timeless art where the earth meets the sky and the sun kisses the soil” [4]. Overall, interacting with plants positively correlates drastically with physical and mental health. As such, leveraging a Plant Care System allows consumers to gain confidence and pride in successfully nurturing their plants.

1.1.2. Customer Base Attributes

The demographic for Plant Care System is young adults, aged 18 to 26 (*Age*). Geographically, we are targeting individuals in Waterloo (*Geographic*). Analyzing the population density, there are 32,675 individuals aged 15 to 19 years; 38,980 individuals aged 20 to 24 years; and 36,935 individuals aged 25 to 29 years [5]. As per the climate, the plant is meant to remain indoors (*Geographic*). Factors to account for in this climate include indoor lighting (natural and artificial light), indoor temperatures, and humidity levels [6]. For the purchasing power, the cost of the Golden Pothos Plant retails for \$15.97 on Home Depot [7]. Furthermore, we seek that the employment status of our users be full-time (*Employment*).

1.1.3. Customer Challenges

Customer challenges include navigating watering, soil, light and temperature changes because plants tend to wilt or grow at stunted rates. Furthermore, users will have to maintain the growing conditions of the plant indoors. Since Waterloo has seasonal climate challenges that vastly differ, balancing the nature of light, temperature, and humidity during colder months of the year can pose a struggle. For instance, in July, the hottest month of the year, the average high is 26 °C and low is 15 °C. However, in January, the coldest month of the year, the average low is -10 °C and high is -3 °C [8]. Additionally, users lack knowledge of the nuances of plant care. Understanding watering schedules, which soil to use, and signs of

plant wilting can be difficult. Dedicating time for consistent care amongst other commitments may mean a decrease in plant care quality [2].

1.2. Competitive Landscape

1.2.1. Waterloo Gardeners - Waterloo Horticultural Society (Social)

Waterloo Gardeners is a volunteer, non-profit organization in Waterloo, Ontario. It is a community group interested in gardening that offers workshops and other interactive activities to encourage the community to get involved.

How it Addresses the Challenges:

The events that Waterloo Gardeners hosts offer in-person opportunities for community individuals to learn more about plant care, soil levels, and gardening mannerisms. This allows individuals to hone in on gardening skills in a social environment that fosters bonds and encourages shared growth.

Shortcomings:

While the workshops are beneficial, access to them is limited to members of the gardening club. Thus, the broader community does not have access to these resources. Furthermore, the large client base of workshops restricts personal guidance and individualized care to address specific gardening issues. Additionally, the volunteer-based nature of the organization leads to inconsistent resource availability among members [9].

1.2.2. Antonki Soil Moisture Meters (Economic)

The Antonki Soil Moisture Meters is a meter that monitors soil moisture level and communicates that data to users. By removing the probe protector and inserting it near the root of the plant, users can quickly read the soil's moisture level and adjust their watering schedule as such.

How it Addresses the Challenge:

The Antonki Soil Moisture Meters have a user-friendly design and clear indicators that allow users to understand their plant's moisture level. It is waterproof for indoor and outdoor conditions, does not require batteries, and is also equipped with a water guide map so users can determine which house zone their plants should be located in.

Shortcomings:

It is evident through the Antonki Soil Moisture Meters that there are numerous soil moisture sensors and pH sensors on the market. However, there are not many plant monitoring systems that monitor and track light levels, temperature, and humidity data over time, as our product does. Ultimately, the Antonki product lacks the functionality and user connection that our plant care system employs [10].

1.2.3. PlantAI: Identifier & Diagnose (Technical)

PlantAI is an AI plant care assistant that responds to plant-related inquiries instantly. It also identifies unknown plants, diagnoses them and includes professional plant care resources.

How it Addresses the Challenge:

PlantAI is an accessible tool that allows users to instantly access information through a photo or scan. As such, it is convenient for individuals with access to a mobile device and provides those without gardening knowledge with quick solutions. Additionally, it produces immediate recommendations for plant care and health preservation, on aspects like watering, sunlight, and soil.

Shortcomings:

Usage of the tool is reliant on a phone or mobile device, thus restricting those who are not familiar with technology or those who do not have access to it. Inconsistent internet access is also a contributing factor. Furthermore, the AI's capability to diagnose complex issues using a photo is limited, in comparison to a human-handled tool [11].

1.3. Requirement Specification

1. The system shall periodically monitor the ambient humidity level using a humidity sensor and alert the user when the humidity rises above 70% or drops below 40% [12][13].
2. The system shall periodically monitor the ambient temperature level using a thermometer and alert the user when the temperature rises above 29°C or drops below 15°C [14].
3. The system shall periodically monitor the light intensity using a light sensor and alert the user when the light intensity rises above 20,000 or drops below 10,000 lux [15].
4. The system shall log sensor data (moisture, temperature, humidity, and light intensity) every 5 minutes and display the data on a 4-digit LED display for user reference [16].
5. The system shall be connected to temperature, humidity, and light intensity sensors using the digital pins on an STM32 microcontroller to gather environmental data every 5 minutes [16].
6. The system shall be powered by a hardwire 12V DC power source [16].
7. The system shall store sensor data in the microcontroller storage drive and have a capacity of 512 KB to store at least 1.5 days of sensor readings at 5-minute intervals [16].
8. The system shall be connected to a 4-digit LED display, digital push button, rotary encoder, white LED, RGB LED, and traffic light sensor using the digital pins on an STM32 microcontroller to display user information and to receive user input whenever this system is turned on via the digital push button [16].
9. All electronic components, including sensors and wiring, shall be housed in IP65-rated waterproof enclosures and properly insulated to keep moisture out, reducing the risk of short circuits or electrical shock when exposed to water during irrigation by the user [17].

2. Analysis

2.1. Design

2.1.1. Physical Schematic

The high-level overview of the design is shown below. This outlines how the Nucleo F401RE's will interact with their respective sensors and displays. It also shows how the two boards will communicate amongst themselves.

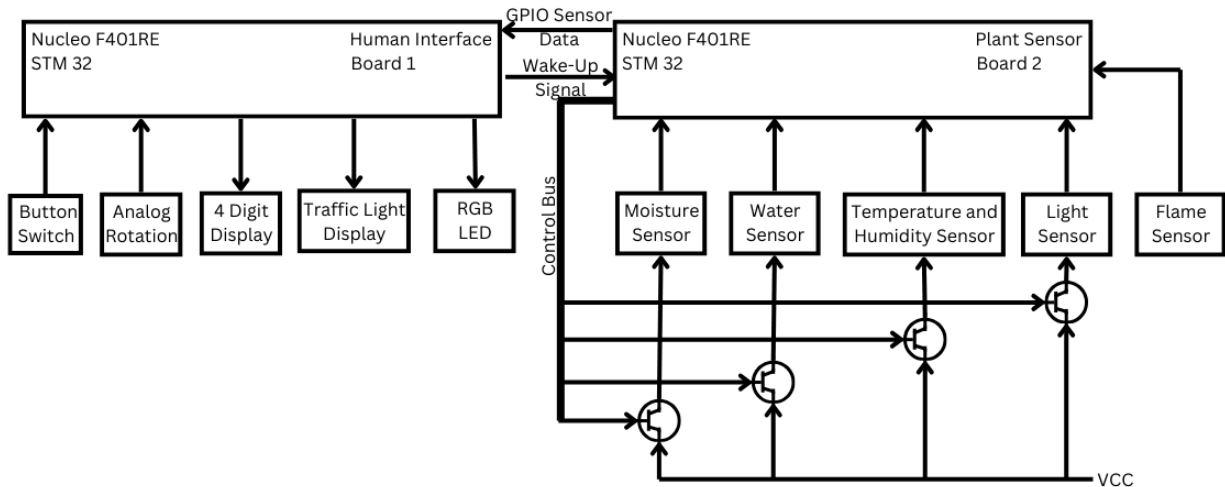


Figure 1: Physical Schematic demonstrating how the Nucleo F401RE interacts with the sensors and displays, and how the two boards communicate.

Legend	
Button Switch	• Used to turn the displays on and off to conserve power and lengthen the lifespan of the displays • Also used to select different options in combination with the rotary encoder
Analog Rotation	• Used to cycle through different options in the 4-digit display and adjust settings
4-Digit Display	• Used to output different sensor readings from the plant • Used to adjust settings such as the sensor sample time
Traffic Light Display	• Has a red, yellow, and green LED • Used to communicate the status of the plant to the user ○ Green indicates that the plant is in good status ○ Yellow indicates that the plant needs help (such as watering) ○ Red indicates that the plant is in severe need of help

RGB LED	● Used to indicate the user interface is up and running ● It is in conjunction with the settings options
Wake-Up Signal	● An on or off signal that can be turned on to “wake up” board 2 ● This results in board 2 measuring all the sensor data and then sending the status to board 1 (through the GPIO sensor Data)
GPIO Sensor Data	● This single data line GPIO communicates the “wake up” sensor data from board 2 to board 1, for board 1’s usage ● Will use USB design communication protocol for this application
Control Bus	● Bus of 4 wires coming from the Nucleo F401RE Board 2 to 4 transistors ● Used to turn on an individual sensor to measure the plant, then turn off the sensor (to conserve energy when constant monitoring status is not needed) ○ The plant’s status needs to be monitored at least once every half an hour, but this can be adjusted by the user
Moisture Sensor	● Will measure the moisture in the soil and take multiple samples over some time ○ This indicates how much water the plant is taking out of the soil so users can determine the best time to water the plant
Water Sensor	● Used to measure large amounts of water in the soil ● If the plant is overwatered, it will be clear from this sensor ● Also used to indicate when the plant was watered so data can be collected
Temperature and Humidity Sensor	● Used to measure both the temperature and humidity from around the plant so data can be collected and plotted
Light Sensor	● Used to give the user a rough estimate of how much sunlight the plant is exposed to
Flame Sensor	● Used to determine if the plant, or if something near the plant is on fire ● If so, it will change the colour of the traffic light to red and turn off all the circuitry so the device does not add to the fire

2.1.2. Circuit Schematic

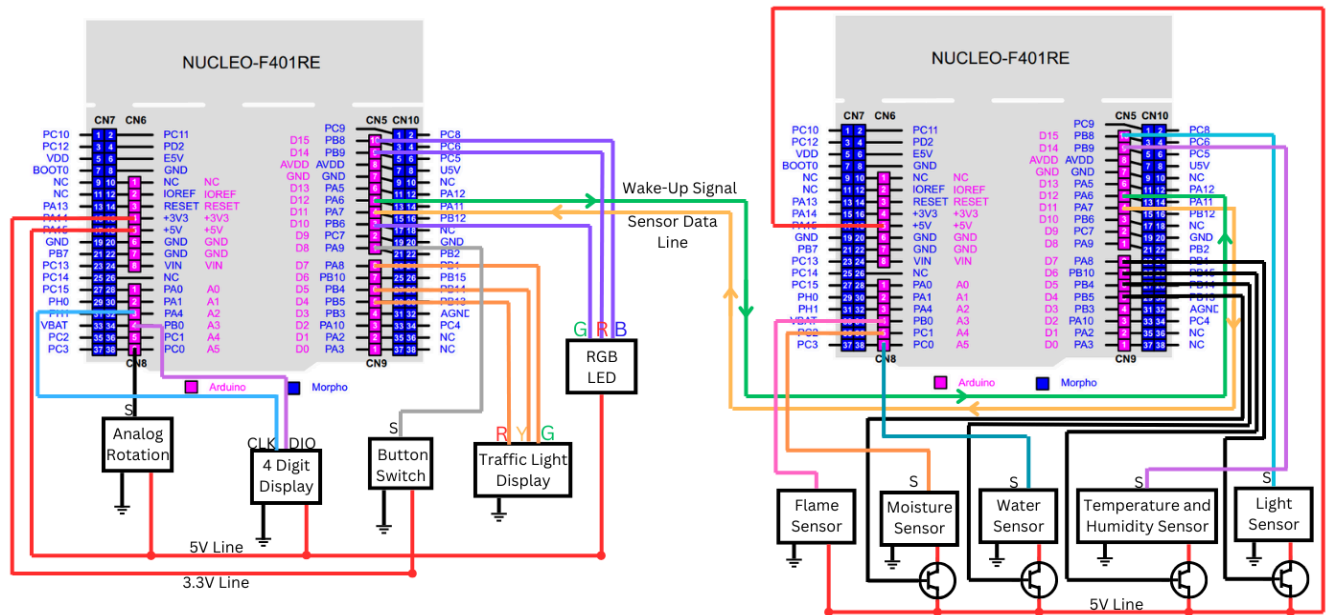


Figure 2: Circuit schematic demonstrating the connections between each of the components on the Breadboard.

2.1.3. Code Flowchart

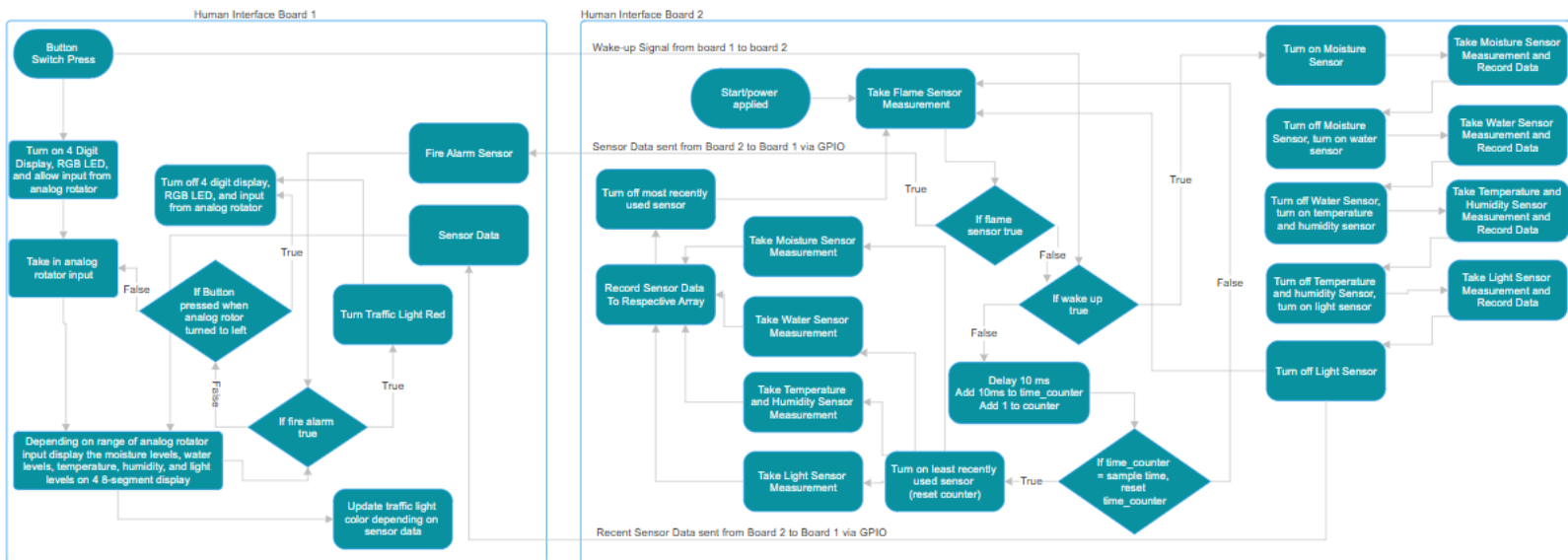


Figure 3: Code chart demonstrating what inputs the product will take in, what decisions it will perform based on this data, and what the corresponding output will be.

2.1.4. Design Alternatives

A morph chart was used in the decision process.

Design Alternatives Through Morph Chart			
Number of Boards	1	2	3
Board Connection	Tx/Rx Communication	GPIO	Specific Wakeup line
Sensors	Moisture sensor	Water sensor	Temperature/Humidity Sensor
	Light sensor	Flame sensor	Analog temp sensor - Too complicated
	Steam sensor		
Input	Touch Module	Button Switch	Rotary Encoder Sensor
	Joystick	Potentiometer	
Output	4-Digit Display	Buzzer	Traffic Light Display
	RGB Light	White LED	Microphone
Power Mitigation	Transistors used to control power to sensors	Use of a battery to supply power to sensors to mitigate large power surges	

Note: Gray indicates that the option was selected for use in the project. White indicates that it was considered but not used.

A computational matrix was used to determine which components above to use.

<i>Scale:</i>
Low - 1
Moderate - 2
High - 3

Component	Cost	Complexity	Incompatibility	Power Inefficiency	Infeasibility	Sum	Decision
<i>Moisture Sensor</i>	1	1	3	2	2	9	Yes
<i>Water Sensor</i>	2	2	2	3	2	11	Yes

<i>Flame Sensor</i>	1	1	1	1	2	6	Yes
<i>Temperature/Humidity</i>	2	2	1	2	2	9	Yes
Analog Temp Sensor	3	3	3	1	3	13	No
Stream Sensor	3	3	3	3	3	15	No

Inputs	Cost	Complexity	Incompatibility	Power Inefficiency	Infeasibility	Sum	Decision
<i>Button Switch</i>	1	2	2	1	1	8	Yes
<i>Rotary Encoder Sensor</i>	2	2	2	1	1	8	Yes
<i>Potentiometer</i>	1	1	2	2	1	7	Yes
Touch Module	2	2	2	2	3	11	No
Joystick	2	1	3	1	3	10	No

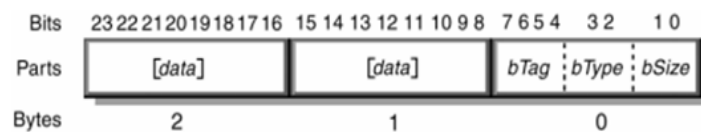
Outputs	Cost	Complexity	Incompatibility	Power Inefficiency	Infeasibility	Sum	Decision
<i>4-Digit Display</i>	1	2	1	1	1	6	Yes
<i>RGB Light</i>	2	1	2	2	1	8	Yes
<i>Traffic Light Display</i>	1	2	1	1	2	7	Yes
Buzzer	2	2	2	1	2	9	No
Microphone	2	3	3	2	2	12	No
White LED	2	3	3	2	3	13	No

2.2. Scientific/Mathematical Principles

Principle #1:

The entire project relies on correctly setting up the plant sensor with the STM32 microcontroller, allowing it to communicate with the user interface STM32 microcontroller through a direct wired connection. This

“communication” occurs over binary data transmission using USB (Universal Serial Bus) protocol. This binary data transmission is a standard we are going to use, and it works via sending packets of binary information back and forth between the two STM32 devices. A packet is 24 bits long containing 2 bytes (8 bits = 1 byte) of data and 1 byte of packet information. This communication standard is found in section 6.2.2.1 of the USB handbook. Using this information, we will be able to communicate between devices through serial protocols [18]. This scientific communication principle will contribute to the whole project by allowing the two STM32 boards to communicate effectively, and efficiently to relay information back and forth.



Application:

1. The application of the binary data transmission via the USB protocol is demonstrated by incorporating a hardwire connection between two STM32 microcontroller setups.
2. This setup allows the plant sensor STM32 microcontroller to transmit sensor data to the user interface STM32 microcontroller (board 1).
3. This then displays the plant's status on a 4-digit LED display and a traffic light indicator, keeping the user informed on the status of the plant.
4. The USB connection will support calibration settings for sensors through the user's computer.
5. This would allow the end user to adjust sensitivity or thresholds depending on the type of plant that they need to take care of.
6. The USB connection will support the transmission of logged data from the STM32 onboard storage to the user's computer.
7. Therefore, the user can graph out trends and determine if anything in the system or surrounding environment is in need of adjustment.
8. The USB protocol will facilitate firmware updates, allowing new functionalities, sensor recalibrations, or bug fixes without needing to replace hardware components.
9. The USB connection will also serve as a power source, powering the entire care system with a hardware power connection from a nearby electrical outlet.
10. If a sensor or another component of the system malfunctions or provides readings outside expected ranges, the USB protocol will relay error codes or diagnostic information to the user interface.
11. This way, the user can troubleshoot directly from the interface without needing direct access to each sensor [18].

Principle #2:

The second principle used is the mathematical principle of modeling voltage (V), current (I), resistance (R), and power (P). This principle will need to be applied to:

- a) Meet project requirements by limiting power usage to ≤ 500 mJ of power at any one instant.
- b) Calculate the appropriate resistors needed to safely use all components so they are not overpowered and can function properly.

The two formulas that will be used to calculate these values are shown below. Formula 1 is for calculating power and Formula 2 is for calculating resistance [19].

Formula 1: $P = I * V$

Formula 2: $\frac{V}{I} = R$

Application:

1. The application of these two formulaic principles will be used to calculate the maximum possible power consumption of the device during the most power-intensive point of the run.
2. Get the voltage that the device takes in [20] and multiply it by the maximum current the device takes in [20] using Formula 1. Thus, get the power draw.
3. If the maximum draw current is unknown, get the voltage that the device takes in [20] and measure the resistance.
4. Afterwards, find the current through Formula 2. Once the current is known, use Formula 1 to get the power.
5. The components' maximum power consumptions are the power consumptions combined and summed together in a logical manner to determine the largest power draw, depending on the sensors that are turned on. This power value comes out to be 10.57 Watts [20].
6. There is no stored energy in this project as there are no batteries, capacitors, or devices which have a specific.
7. These formulas will also be used to calculate the maximum power that an outlet adapter (from the wall outlet to USB) will need to be able to handle.
8. This value will be the maximum power draw that the system will use, which is 10.57 Watts. Thus, the wall adapter will have to be capable of handling 10.57 watts.
9. Adding onto the previous point, the wall outlet will need to be able to handle 10.57 Watts because this is where the energy comes from.
10. At any one point, the board's components may need to draw a lot of wattage. Thus, we need to ensure that each voltage output of the board is less than the maximum output wattage, which is 0.20 watts per port [20]. Since this is greater than the input of any sensor, the board will be able to supply the required amount of power.
11. This above point also means that the wires will need to be able to support 0.20 watts. The wires used in this project will easily be able to handle this as they are rated for 3.5 amps [20].

Principle #3:

The third principle that will be applied is the use of Boolean logic. All the components that are connected to the boards communicate to the board via binary strings. To use and access these strings, the code will perform Boolean logic operators to get the information from the strings. An example of this is the humidity/temperature sensor.

0111 0100 1001 0110	Sensor Input via binary string
0000 0000 1111 0000	Selection Boolean statement (Mask)
AND = 0000 0000 1001 0000	

Application:

1. This sensor communicates both the humidity and the temperature data in one binary string to get the specific data, which is made up of many bytes.
2. We need to apply the boolean logic operator & (AND) with a string of 0's and a string of 1's to access specific bytes.
3. This will read either the temperature data or the sensor input from the sensor.
4. Once the byte data is read, it will be bit-shifted right until it is at the start of the string.
5. Here, the values will be converted from binary to decimal, which will allow the user to read the temperature and humidity data.
6. When it comes to the communication of the boards, the communication line between board 1 and board 2 will also be sent through binary transmission.
7. The second board will send a binary string to the first. This will also have to be decoded via the AND operator and a binary mask of 1's, where we need the data, and 0's where we don't.
8. This string of bits will then be bit shifted to the right until it is the first byte in the string.
9. Similarly, this decided data will be converted into decimal form which can then be displayed to the user.
10. The usage of the board between the user and the human interface board will be controlled by a binary switch. When this switch is on ("1"), the board will turn on, and when it is on ("0"), the board will be off [20].

3. Costs

3.1. Manufacturing Costs

Components

- STM32F401RE Nucleo-Board (x2) - $(\$46.10 + \text{tax}) \times 2 = \104.18
 - Manufacturer: STMicroelectronics
 - Main Address: Chemin du Champ-des-Filles 39 Plan-les-Ouates, 1228, Geneva, Switzerland
 - Distributor: W Store - Address: 200 University Ave West, Waterloo, ON N2L
- KEYESTUDIO 37 in 1 Sensors Starter Kit - $\$38.00 + \text{tax} = \42.94
 - Manufacturer: Keyes
 - Main Address: Shenzhen, Guangdong 518109, CN
 - Distributor: Amazon - Address: 410 Terry Ave N, Seattle, WA, 98109, USA [22]
- 2N3906 Single PNP Bipolar Transistor - $\$1.04 + \text{tax} = \1.18
 - Manufacturer: STMicroelectronics

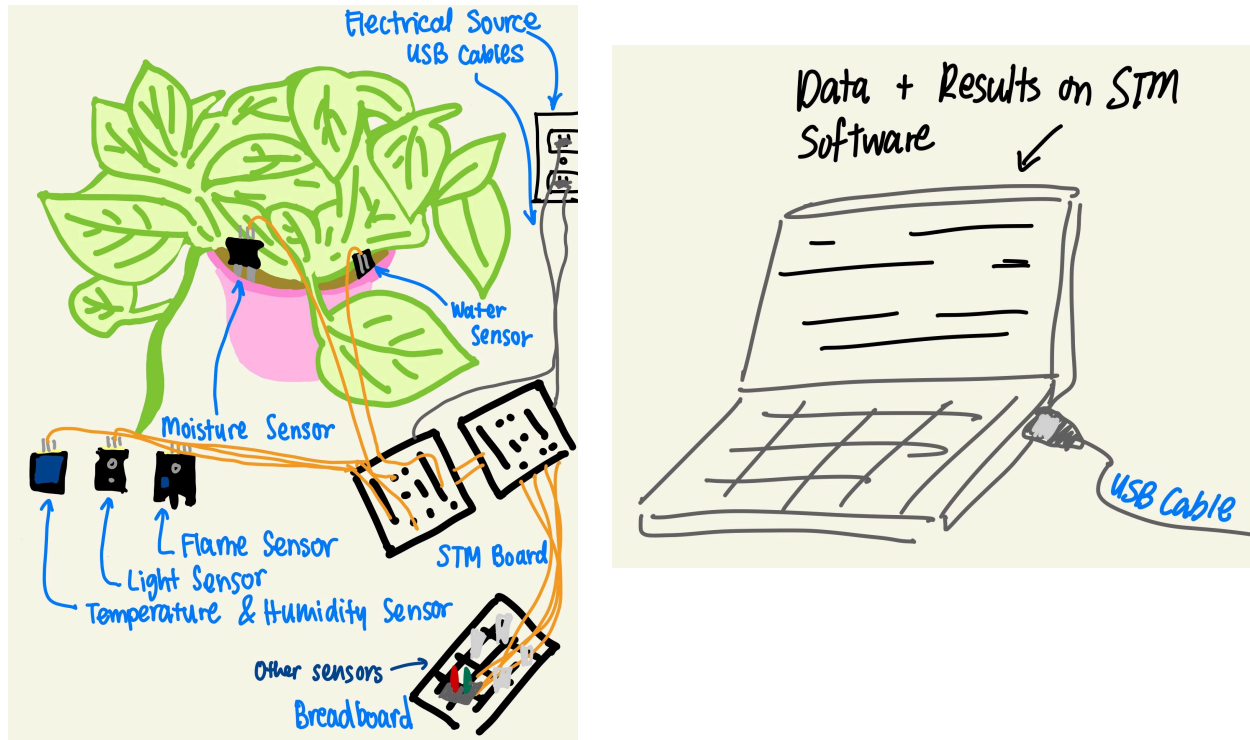
- Main Address: 350 Burnhamthorpe Rd W, Mississauga, ON L5B 3J1
- Distributor: Digikey: Address: 701 Brooks Avenue South, Thief River Falls, MN, 56701, USA [23]
- 3M Multi-Use Colored Duct Tape, Black - $\$6.47 + \text{tax} = \7.31
 - Manufacturer: Scotch
 - Main Address: 300 Tartan Drive. London, Ontario. N5V 4M9
 - Distributor: Amazon - Address: 410 Terry Ave N, Seattle, WA, 98109, USA [24]
- BB830 Solderless BreadBoard, White - $\$10.65 + \text{tax} = \12.03
 - Manufacturer: BusBoard Prototype Systems
 - Main Address: #612-500 Country Hills Blvd NE, Suite #140 Calgary, AB T3K 5K3
 - Distributor: Amazon - Address: 410 Terry Ave N, Seattle, WA, 98109, USA [25]

3.2. Implementation Costs

3.2.1. Additional User Costs

- Golden Pothos Indoor Vining Houseplant in 6 in. Grower Pot - $\$15.97 + \text{tax} = \18.05
 - Manufacturer: United Nursery
 - Main Address: 30401 SW 217th Ave, Homestead, FL 33030, United States
 - Distributor: 600 King St N, Waterloo, ON N2V 2J5 [7]

3.2.2. Installation Manual



Figures 4 & 5: Visual demonstration of the installation manual and set up of the product.

1. Take the Plant Care System out of the box.
2. Place the Golden Pothos Plant upright and near an electrical source to ensure the Plant Care System can start operating.
3. Ensure that the following sensors are in their corresponding locations:
 - a. Moisture and water → In the plant soil
 - b. Light, temperature, humidity, and flame → External to the plant
4. Plug in the STM32 Nucleo-Board's USB cable into an electrical source.
5. Ensure that the button's pins remain secure within the breadboard sockets.
6. Press the "Reset" button on both boards to start the system.
7. After momentarily unplugging the system, plug the USB into a computer. Access a text document on the STM32 software verifying the Plant Care System's data and results.

3.2.3. User Guide

1. Place the Golden Pathos Plant upright and near an electrical outlet.
2. Ensure that the sensors are positioned as such:
 - a. Moisture and water → In the plant soil
 - b. Light, temperature, humidity, and flame → External to the plant
3. Plug in the USB cable of the STM32 Nucleo-Board into the electrical outlet
4. *Moisture Monitoring:* The system will continuously monitor the moisture levels of the soil. Water the plant as needed according to these results.

5. *Light Monitoring:* The system will continuously monitor the light levels of the plant. Adjust the lighting exposure of the plant as needed according to these results.
6. *Temperature and Humidity Monitoring:* The system will continuously monitor the temperature and humidity levels of the plant. Adjust the temperature and humidity range that the plant is exposed to if it falls out of the ideal range.
7. *Flame Detection:* If a flame is detected by the system, it will immediately inform the user.
8. *Maintenance:* Routinely check the sensors to ensure their cleanliness and security.
9. *Power off:* Unplug the USB from the system when it is not being used.

4. Risks

4.1. Energy Analysis

(Max possible values) (All images from [20])

4.1.1. STM32 NUCLEO-F401RE (x2)

4.1.1.1. Current draw 100 mA [26]

4.1.1.2. Voltage draw 5V [27]

4.1.1.3. Power = 10 Watts

Specifications		
Product Attribute	Attribute Value	Select Attribute
Manufacturer:	STMicroelectronics	☐
Product Category:	Development Boards & Kits - ARM	
RoHS:	 Details	
Series:	NUCLEO-F401RE	☐
Product:	Development Boards	☐
Core:	ARM Cortex M4	☐
Tool Is For Evaluation Of:	STM32F401RET6	☐
Packaging:	Bulk	☐
Brand:	STMicroelectronics	
Data Bus Width:	32 bit	
Description/Function:	STM32 nucleo board	
For Use With:	STM32F	
Interface Type:	USB	
Operating Supply Voltage:	3.3 V, 5 V, 7 V to 12 V	
Product Type:	Development Boards & Kits - ARM	
Factory Pack Quantity:	1	
Subcategory:	Development Tools	
Tradename:	NUCLEO	
Unit Weight:	385.560 g	

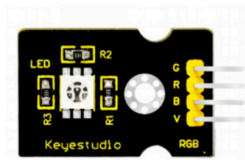
4.1.2. RGB LED

4.1.2.1. Voltage 5V [20]

4.1.2.2. Resistance 98 ohms

4.1.2.3. Power = 0.255 Watts

Project 2: RGB LED



Description

This is a full-color LED module, which contains 3 basic colors - red, green and blue. They can be seen as separate LED lights. After programming, you can turn them on and off by sequence or can also use PWM analog output to mix three colors to generate different colors.

Specification

- Color: red, green and blue
- Brightness: High
- Voltage: 5V
- Input: digital level
- Size: 30*20mm
- Weight: 3g

4.1.3. Traffic Light

4.1.3.1. Voltage 3.3V [20]

4.1.3.2. Resistance 103 ohms

4.1.3.3. Power 0.106 Watts

Project 3: Traffic Light



Description

When learning the microcontroller, you may usually use three LEDs, namely red, green and yellow lights to simulate the traffic light blinking via external connection.

This time we specially design this module which is very convenient for wiring, and on the module you can see the red, yellow and green LED. This module is fully compatible with Arduino microcontroller and Raspberry Pi system.

Specification

- Working Voltage: 3.3-5v
- Interface Type: digital
- PH2.54 Socket

4.1.4. Digital Push

4.1.4.1. Voltage 3.3V [20]

4.1.4.2. Resistance 152 ohms

4.1.4.3. Power 0.072 Watts

Project 7: Digital Push Button



Introduction:

This is a basic button module. You can simply plug it into an IO shield to have your first try of Arduino. **Features:**

Wide voltage range from 3.3V to 5V Easily recognizable interfaces of sensors ("A" for Analog and "D" for Digital) Standard assembled hole Clear icons illustration High quality connector Easy to plug and operate Large button and high-quality cap To achieve interesting and interactive works **Specification:**

- Supply Voltage: 3.3V to 5V
- Interface: Digital
- Size: 30*20mm
- Weight: 4g

4.1.5. Flame Sensor

4.1.5.1. Voltage 5V [20]

4.1.5.2. Resistance 10k ohms

4.1.5.3. Power 0.003 Watts

Project 16: Flame Alarm



Description:

This flame sensor can be used to detect fire or other lights with wavelength stands at 760 nm ~ 1100 nm. In the fire-fighting robot game, the flame plays an important role in the probing, which can be used as the robot's eyes to find fire source.

Specification:

- Supply Voltage: 3.3V to 5V
- Detection range: 20cm (4.8V) ~ 100cm (1V)
- Rang of Spectral Bandwidth: 760nm to 1100nm
- Operating temperature: -25°C to 85°C
- Interface: digital
- Size: 44*16.7mm
- Weight: 4g

4.1.6. Analog Rotation

4.1.6.1. Voltage 5V [20]

4.1.6.2. Resistance 10k ohms

4.1.6.3. Power 0.003 Watts

Project 20: Analog Rotation



Description:

This analog rotation sensor is Arduino compatible. It is based on a potentiometer. Its voltage can be subdivided into 1024, easy to be connected to Arduino with our sensor shield. Combined with other sensors, you can use it to make interesting projects by reading the analog value from the IO port.

Specification:

Supply Voltage: 3.3V to 5V Interface: Analog Size: 30*20mm Weight: 8g

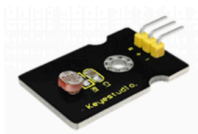
4.1.7. Light Sensor

4.1.7.1. Voltage 5V [20]

4.1.7.2. Resistance 50k ohms

4.1.7.3. Power 0.0002 Watts

Project 21: Photocell



Description:

Photocell is commonly seen in our daily life and is mainly used in intelligent switch, also in common electronic design. To make it easier and more effective, we supply the corresponding modules. Photocell is a semiconductor. It has features of high sensitivity, quick response, spectral characteristic and R-value consistence, maintaining high stability and reliability in environment extremely such as high temperature and high humidity. It's widely used in automatic control switch fields like cameras, garden solar lights, lawn lamps, money detectors, quartz clocks, music cups, gift boxes, mini night lights, sound and light control switches, etc.

Specification:

- Interface type: analog
- Working voltage: 5V
- Size: 30*20mm
- Weight: 3g

4.1.8. Water Level

4.1.8.1. Voltage 5V [20]

4.1.8.2. Max Current 20mA [20]

4.1.8.3. Power 0.1 Watts

Project 23: Water Level



Description:

Keystudio water sensor is easy- to-use, portable and cost-effective, designed to identify and detect water level and water drop. This smaller sensor can measure the volume of water drop or water quantity through an array of traces of exposed parallel wires.

Features:

smooth conversion between water quantity and analog quantity; strong flexibility, outputting basic analog value; low power consumption and high sensitivity; directly connect to microprocessor or other logic circuits, suitable for a variety of development boards and controllers such as Arduino controller, STC single-chip microcomputer, AVR single-chip microcomputer and more.

Specification:

- Operating voltage: DC5V
- Operating current: < 20mA
- Sensor type: Analog
- Detection area: 40mm x16mm
- Production process: FR4 double-side tinned
- Shape design: Anti-skid semi-lunar recess
- Working Temperature: 10°C-30°C
- Working Humidity: 10%-90% without condensation
- Weight: 3g
- Dimensions: 65mm x 20mm x 8mm

4.1.9. Moisture Sensor

4.1.9.1. Voltage 5V [20]

4.1.9.2. Max Current 20mA [20]

4.1.9.3. Power 0.1 Watts

Project 24: Soil Moisture



Description:

This is a simple soil humidity sensor aimed to detect the soil humidity. If the soil is in lack of water, the analog value output by the sensor will decrease, otherwise, it will increase. If you use this sensor to make an automatic watering device, it can detect whether your botany is thirsty to prevent it from withering when you go out. Combine this sensor with Arduino controller can make your plant more comfortable and your garden more smarter. The soil humidity sensor module is not as complicated as you might think, so if you need to detect the soil in your project, it will be your best choice. The sensor is set with two probes which are inserted into the soil. If the current goes through the soil, the sensor will get resistance value by reading the current changes between the two probes, then convert the resistance value into moisture content. The higher moisture (less resistance), the higher conductivity the soil has. The surface of the sensor has undergone metallization process to prolong its service life. Insert it into the soil and then use the AD converter to read it. With the help of this sensor, the plant can remind of you: I need water.

Specification:

- Power Supply Voltage: 3.3V or 5V
- Working Current: ≤ 20mA
- Output Voltage: 0-2.3V (When the sensor is totally immersed in water, the voltage will be 2.3V) The higher humidity, the higher the output voltage.
- Sensor type: Analog output
- Interface definition: Pin1- signal, Pin2- GND, Pin3 - VCC
- Service life: About one year (gold-plated surface for enhancing conductivity and corrosion resistance)
- Module size: 20*60mm

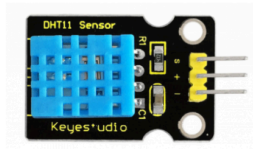
4.1.10. Temperature/Humidity Sensor

4.1.10.1. Voltage 5V [20]

4.1.10.2. Resistance 5k ohms

4.1.10.3. Power 0.005 Watts

Project 35: Temperature and Humidity Display



Introduction:

This DHT11 sensor features calibrated digital signal output with the temperature and humidity sensor complex. Its technology ensures high reliability and excellent long-term stability. A high-performance 8-bit microcontroller is connected on the sensor. This sensor includes a resistive element and a sense of wet NTC temperature measuring devices. It has advantages of excellent quality, fast response, anti-interference ability and high cost performance. Each DHT11 sensor features extremely accurate calibration data of humidity calibration chamber. The calibration coefficients stored in the OTP program memory, internal sensors detect signals in the process, and we should call these calibration coefficients. The single-wire serial interface system is integrated to make it quick and easy. Qualities of small size, low power, and 20-meter signal transmission distance make it a wide applied application or even the most demanding one.

Specification:

- Supply Voltage: +5 V
- Temperature range: 0-50 °C error of ± 2 °C
- Humidity: 20-90% RH $\pm 5\%$ RH error
- Interface: Digital
- Size: 30*20mm
- Weight: 4g

4.1.11. 4-Digit LED

4.1.11.1. Voltage 5V [20]

4.1.11.2. Resistance 812 ohms

4.1.11.3. Power 0.031 Watts

Project 37: 4-digit LED Display

Description:

The keyestudio 4-digit LED Display Module integrates a 0.36" 4-Digit 7-Segment Display Common Anode with 12 pins. It uses the driver chip TM1637. The module has 4 control pins of 2.54mm pitch, which can direct connect to microcontroller with jumper wires. So it's very easy to control the matrix without tons of wiring. Available to make microcontroller control the 4-digit LED segment display via signal interface, greatly saving IO pin resource on microcontroller. The module has two 3mm fixed holes, convenient for mounting on other devices. If you've been eyeing matrix displays but hesitated because of the complexity, this is the solution you've been looking for!



Technical Details:

- Operating voltage: DC 5V
- Operating temperature range: -40 ~ +85°C
- Dimensions: 45mm*21mm*10mm
- Weight: 7.2g

The total power usage is less than 30 Watts.

4.2. Risk Analysis

Risks associated with users using the system as intended include:

1. The various sensors can detect the wrong values, possibly leading to the plant withering and dying.
2. The software running the system can potentially go haywire, and interpret the values received from the sensors incorrectly, possibly leading to the plant withering and dying.
3. If the waterproofing of the system fails, this can lead to short circuits or electrical shock when exposed to water during irrigation by the user or simply by being exposed to lots of moisture.

Risks associated with users using the system incorrectly include:

1. If the various sensors are placed too far from the plant, this can lead to inaccurate values being measured.

2. If the system and the plant are placed outdoors, the rapid humidity, temperature, and light intensity changes may lead to the system wearing out faster than intended.
3. If the user does not regularly maintain the system, including cleaning the sensors, the sensors may start to record inaccurate values.

Risks associated with users misusing the system include:

1. If the system is used to measure anything extreme, such as extremely hot temperatures or beams of concentrated, bright light, the sensors may get damaged, and possibly break down and release harmful materials into the environment.
2. If the user changes the warning cutoffs (for humidity, temperature, and light intensity) to values that are outside of the recommended ranges, this may lead to an excessive or insufficient supply of water, heat, or light for the plant.
3. The user may mishandle the delicate electrical components, leading to an increased risk of electrical shock, inaccurate sensor readings, and wearing out the system faster than intended.

Possible ways that the design could malfunction and their consequences on safety or the environment:

1. If the system is exposed to moisture over a long time, this may lead to corrosion of the electrical components and may lead to leaking harmful chemicals into the soil, as well as needing to replace the components more, contributing to electronic waste.
2. If the sensors are not calibrated correctly, this may lead to increased water and energy use, contributing to environmental strain.
3. Voltage fluctuations and/or power interruptions may lead to system instability and damage to the components of the system, contributing to electronic waste.

5. Testing and Validation

5.1. Test 1

Test that the humidity measurement occurs every 5 minutes and displays on the 4-digit LED display.

- Setup: The STM32 board is connected to a laptop, and the humidity sensor is connected to the STM32 board.
- Environmental parameters: The humidity of the testing room should be the default, i.e., with the windows closed and the AC off.
- Inputs: An action command is given to the STM32 to start taking humidity data. The stop command will be given later.
- Quantifiable measurement standard: the humidity around the sensor.
- Pass criteria: the humidity sensor updates at 5-minute intervals (with the humidity around the sensor changing).

5.2. Test 2

Test that the temperature measurement occurs every 5 minutes and displays on the 4-digit LED display.

- Setup: The STM32 board is connected to a laptop, the temperature sensor is connected to the STM32 board.
- Environmental parameters: The testing room should be at room temperature, i.e., with the windows closed and the AC off.
- Inputs: An action command is given to the STM32 to start taking temperature data. The stop command will be given later.
- Quantifiable measurement standard: the temperature around the sensor.
- Pass criteria: the temperature sensor updates at 5-minute intervals (with the temperature around the sensor changing).

5.3. Test 3

Test that the light intensity measurement occurs every 5 minutes and displays on the 4-digit LED display.

- Setup: STM32 board is connected to laptop, light intensity sensor is connected to STM32 board.
- Environmental parameters: The light intensity of the testing room should be the default, i.e., with the window shades down and the room light on.
- Inputs: An action command is given to the STM32 to start taking light-intensity data. The stop command will be given later.
- Quantifiable measurement standard: the light intensity around the sensor.
- Pass criteria: the light sensor updates at 5-minute intervals (with the light intensity around the sensor changing).

5.4. Test 4

Test that the IP-65 enclosure keeps solids and liquids out, specifically from all dust and water jets from any direction.

- Setup: The STM32 board is connected to a laptop, STM32 board is enclosed within the enclosure.
- Environmental parameters: N/A
- Inputs: N/A
- Quantifiable measurement standard: the dust and water resistance.
- Pass criteria: the enclosure can keep dust out, as well as water jets from a spray bottle.

5.5. Test 5

Test that the STM32 board can store 1.5 days of sensor data at 5-minute intervals.

- Setup: The STM32 board is connected to a laptop, all sensors are connected to the STM32, system is active and logging data.
- Environmental parameters: The plant care system should all be placed in the same place in the same room throughout the 1.5 days.
- Inputs: An action command is given to the STM32 to start taking all sensor data. The stop command will be given after 1.5 days.
- Quantifiable measurement standard: the amount of data stored.
- Pass criteria: the STM32 board can store 1.5 days of sensor data at 5-minute intervals.

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